Course Code: CS1006 **Name of the Programme: B.Tech (H)** **Name of the Course: Data Structures** **Semester:**		
Course Credit	**No. of hours per week (L + T + P)**	**Total no. of Teaching hours**
4	2-1-2	75
Course Objectives:		
A. To understand the concept of Linear and Non-Linear Data Structures		
B. To solve the real-world problems by applying the Data Structure concept		
C. To evaluate the concept of Data Structures through Programming.		
D. To understand the need for searching, sorting and hashing in the real-world scenario.		

Syllabus:

Module – 1 Abstract Data Types & Memory Allocation	**No. of hours**
	9
Strategies for choosing the appropriate data structure. Abstract data types. Stack Memory - Compile Time, Static memory allocation. Heap Memory - Runtime, dynamic memory allocation. Arrays – Revision of basic operations. Implementation of basic operations – Create, Insert, Traverse, Delete, Search and Update.	

Module – 2 Linked List	**No. of hours**
	9
Linked lists – Types – Singly Linked List – Doubly Linked List – Circular Linked List -(Circular Singly, Circular Doubly). Implementation of operations – Create, Insert, Traverse, Delete, Search and Update.	

Module – 3 Stacks and Queues	**No. of hours**
	9
Introduction to Stacks and Queues. Implementation of their basic operations. Use basic arrays & link list to create stack & queue data structures.	

Module – 4 Sorting, Searching, and Hashing	**No. of hours**
	9
Linear Search, Binary Search, Bubble Sort, Selection Sort, Insertion Sort. Hashing – Hash tables, has functions, strategies to avoid and resolve collisions.	

Module – 5 Trees	**No. of hours**
	9
Binary Tree, Binary Search Tree, AVL Tree, B-Trees, Common operations on these trees such as select min, select max, create, insert, delete. Perform various types of Tree Traversals.	

Course Outcomes
a. Understand and explore the concepts of linear and non-linear data structures to develop solutions for diverse problems.
b. Apply the concepts of hashing, sorting and searching techniques to get an optimized solution.
c. Evaluate and apply appropriate data structure(s) for real-world problems.
d. Demonstrate Good Coding Practices involving lifelong learning.

Text Books	
1	E. Horowitz and S. Sahni, *Fundamentals of Data Structures in C*, 2nd ed. Hyderabad, India: Universities Press, 2014, ISBN: 978-8173716058.
2	R. F. Gilberg and B. A. Forouzan, *Data Structures: A Pseudocode Approach with C*, 2nd ed. New Delhi, India: Cengage Learning India Pvt. Ltd., 2014, ISBN: 978-8131503140.
3	T. L. Kruse, *Data Structures and Program Design in C*, 2nd ed. New Delhi, India: Pearson India, 2001, ISBN: 978-8177584233.

Reference Books	
1	J.-P. Tremblay and P. G. Sorenson, *An Introduction to Data Structures with Applications*, 2nd ed. New Delhi, India: McGraw Hill, 2013.
2	A. M. Tenenbaum, *Data Structures Using C*, New Delhi, India: PHI Learning, 1989.

Lab Programs (30 Hours)	
1	Implementation of Recursion concept
2	Implementation of List using arrays
3	Implementation of Stack and its operations using arrays
4	Implementation of Queue and its operations using arrays
5	Implementation of Linked List and its operations (Singly, Doubly & Circular)
6	Implementation of Linear search and Binary search (both iterative & recursive)
7	Implementation of Insertion Sort
8	Implementation of Hash table using Linear Probing
9	Implementation of Binary search Tree – Creation, Traversal, Searching and Deletion

Tutorials (15 Hours)	
1	Recursion concept
2	List using arrays
3	Stack and its operations using arrays
4	Queue and its operations using arrays
5	Application of Stacks and Queues
6	Singly Linked List and its operations (Singly, Doubly & Circular)
7	Doubly Linked List and its operations
8	Singly Circular Linked List and its operations
9	Doubly Circular Linked List and its operations
10	Linear search and Binary search (both iterative & recursive)
11	Implementation of Insertion Sort, Selection Sort & Bubble Sort
11	Hashing using Linear Probing, Quadratic Probing
12	Hashing using Double Hashing
13	Binary search Tree – Creation, Traversal, Searching and Deletion
14	AVL Tree and its Rotations
15	B-Trees - Creation, Traversal, Searching and Deletion